GORENSTEIN PROPERTY ALONG COMPOSITIONS OF MORPHISMS OF ALGEBRAIC SPACES, WITH DESCENT ALONG FLAT MORPHISMS

THE STACKS PROJECT AUTHORS

Editorial note. Statement span is lines 6611-6628 of sources/stacks-spaces-more-morphisms/pinned-original-source.tex; proof span is lines 6630-6651, the complete author proof (20 body lines).

Editorial note. Navigation stubs: 3 cited-but-not-included labels are supplied as non-load-bearing pointers with their pinned tags where the snapshot tag index has them.

Editorial note. extra_packages is empty on purpose: the retained preamble already loads xy with options and a second declaration clashes in the assembler wrapper.

Editorial note. Text is retained verbatim, including the author's own typos and placeholders.

Allowed standard prerequisites (editorial).

- Explicit prerequisite boundary. The etale-local nature of the Gorenstein property on algebraic spaces, the pointwise ring criterion (lemma-gorenstein, same chapter), and the flat-under-Gorenstein lemma of Dualizing Complexes; cited inputs, with the exhibited homomorphism written out in the retained span.

Source reference. Original lemma 0C11: sources/stacks-spaces-more-morphisms/pinned-original-source.tex, source lines 6651–6651. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 0BJL: sources/stacks-spaces-more-morphisms/pinned-original-source.tex, source lines 6651–6651. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 0E15: sources/stacks-spaces-more-morphisms/pinned-original-source.tex, source lines 6538–6550. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Lemma 0.1. *Let S be a scheme. Let $f : X \to Y$ and $g : Y \to Z$ be morphisms of algebraic spaces over S. Assume that the fibres of f, g, and $g \circ f$ are locally Noetherian. Let $x \in |X|$ with images $y \in |Y|$ and $z \in |Z|$.*

1. *If f is Gorenstein at x and g is Gorenstein at $f(x)$, then $g \circ f$ is Gorenstein at x.*
2. *If f and g are Gorenstein, then $g \circ f$ is Gorenstein.*
3. *If $g \circ f$ is Gorenstein at x and f is flat at x, then f is Gorenstein at x and g is Gorenstein at $f(x)$.*

Date: stacks-project:spaces-more-morphisms.tex:local-snapshot-20261011; GNU FDL 1.2 or later.

(4) *If $f \circ g$ is Gorenstein and f is flat, then f is Gorenstein and g is Gorenstein at every point in the image of f.*

Proof. Working étale locally this follows from the corresponding result for schemes, see Duality for Schemes, Lemma 0C11. Alternatively, we can use the equivalence of (a) and (b) in Lemma 0E15. Thus we consider the local homomorphism of Noetherian local rings

$$\mathcal{O}_{Y,\overline{y}}/\mathfrak{m}_{\overline{z}}\mathcal{O}_{Y,\overline{y}} \longrightarrow \mathcal{O}_{X,\overline{x}}/\mathfrak{m}_{\overline{z}}\mathcal{O}_{X,\overline{x}}$$

whose fibre is

$$\mathcal{O}_{X,\overline{x}}/\mathfrak{m}_{\overline{y}}\mathcal{O}_{X,\overline{x}}$$

and we use Dualizing Complexes, Lemma 0BJL. □